

Statistical Structure in the 2026 ACIC Multi-Treatment Data

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Abstract

We study what 18 randomized data sets from the 2026 ACIC multi-treatment challenge reveal about outcomes and treatment effects. Few category-by-treatment combinations have very small samples, while three numerical variables frequently equal zero. Large mean differences account for most of the total distance between treatment-arm distributions, although mean differences are less prominent in a typical comparison. Covariate adjustment gives smaller estimated standard errors for average effects. We explain how these findings relate to our original submission and why predicting outcomes is different from predicting which treatment works better. Released truth makes this distinction concrete: effects are constant across sampled units within each contrast, but our submission predicts variation.

Keywords: multi-arm trial, exploratory data analysis, outcome distributions, covariate adjustment, conditional average treatment effect

1. Data and questions

We study 18 curated randomized data sets, each with 1,000 units, 40 pretreatment covariates, and five treatments (ACIC Data Challenge, 2026). Arm a is the control. For active arm z , write $m_z(x) = E\{Y(z) \mid X = x\}$ and $\tau_z(x) = m_z(x) - m_a(x)$. The challenge asks for this conditional effect at each sampled unit (iCATE), its sample and X_{12} subgroup averages (sCATE and subCATE), and its population average (PATE).

Our questions concern the data: which covariate values are poorly represented, how outcome distributions differ between arms, and whether covariates help estimate average effects. The original submission provides a modeling example throughout. The analyses were conducted afterward and should not be read as the basis for the original choices.

2. Original submission

The submission first compared linear and tree-based models by cross-validated outcome prediction error. Models with similar held-out predictions were grouped, and the best from each group supplied regressions for S-, T-, X-, and doubly robust learners (Künzel et al., 2019; Kennedy, 2023).

We then compared these learners in local plasmodes: simulated data that retain observed covariates but have known effects (Franklin et al., 2014). A pooled XGBoost outcome model (Chen and Guestrin, 2016), called the creator, supplied conditional means; treatments and residuals were redrawn. Winners with Confidence (Zhang et al., 2024) selected separate sets for effect error and treatment-choice regret. Each set was averaged uniformly. Thus outcome prediction screened the models, while effects supplied by the creator selected the causal learners.

3. Data structure and empirical findings

3.1 Covariate support and variation across data sets

A small treatment group need not imply that many covariate categories are poorly represented. We counted observations in each combination of treatment and category for X_1 – X_6 . Only 24 of 2,340 combinations (1.0%) had fewer than five observations, and 21 involved X_6 . Limited category counts are therefore concentrated in one variable.

A more widespread feature is the frequency of zeros in X_{38} – X_{40} . Between 25.5% and 40.6% of observations equal zero, depending on the variable and data set. Table 1 summarizes these features.

Table 1: Data summaries: means and observed minimum–maximum ranges, not confidence intervals. Arm size summarizes 90 arms; other rows summarize 18 data sets. Ratios compare within-data-set maxima and minima. Small cells have fewer than five observations among 130 category-by-arm cells per data set.

Quantity	Mean	Range
Treatment-arm size	200	112–328
Arm-size ratio	1.89	1.26–2.85
Small category-by-arm cells	1.33	0–3
Zero fraction in X_{38} (%)	32.2	29.7–35.1
Zero fraction in X_{39} (%)	37.4	35.0–40.6
Zero fraction in X_{40} (%)	27.7	25.5–29.9
Outcome variance ratio	1.28	1.05–1.79

Our submission represented categories by indicators. For X_{38} – X_{40} , it used both a zero indicator and a standardized positive component. This allows a model to distinguish being zero from changes among positive values. The observed distributions explain what this encoding represents; they do not show that it predicts better than another encoding.

3.2 Outcome differences between treatment arms

Do treatment arms differ mainly in average outcome, or also in spread and shape? Figure 1A starts with mean differences, $\hat{\Delta}_{dz} = \bar{Y}_{dz} - \bar{Y}_{da}$, for data set d . The 72 estimates range from -1.07 to 4.04 . Twenty pointwise 95% intervals based on Welch (1947) lie above zero; these intervals are not adjusted for multiple comparisons.

Mean differences leave other features of the distributions unexamined. Wasserstein distance compares their quantiles, so it can detect differences even when the means agree. Under randomization and consistency, with positive arm probabilities, the observed distribution in arm j identifies $P_j = \mathcal{L}\{Y(j)\}$. Let μ_j and $\sigma_j > 0$ be its mean and standard deviation, and let P_j° be its centered, standardized distribution. For finite second moments (Irpino and Verde, 2015, Eq. (15)),

$$W_2^2(P_z, P_a) = \underbrace{(\mu_z - \mu_a)^2}_{\text{location}} + \underbrace{(\sigma_z - \sigma_a)^2}_{\text{scale}} + \underbrace{\sigma_z \sigma_a W_2^2(P_z^\circ, P_a^\circ)}_{\text{shape}}.$$

The first two terms measure differences in means and standard deviations. The third measures what remains after each distribution has been centered and standardized.

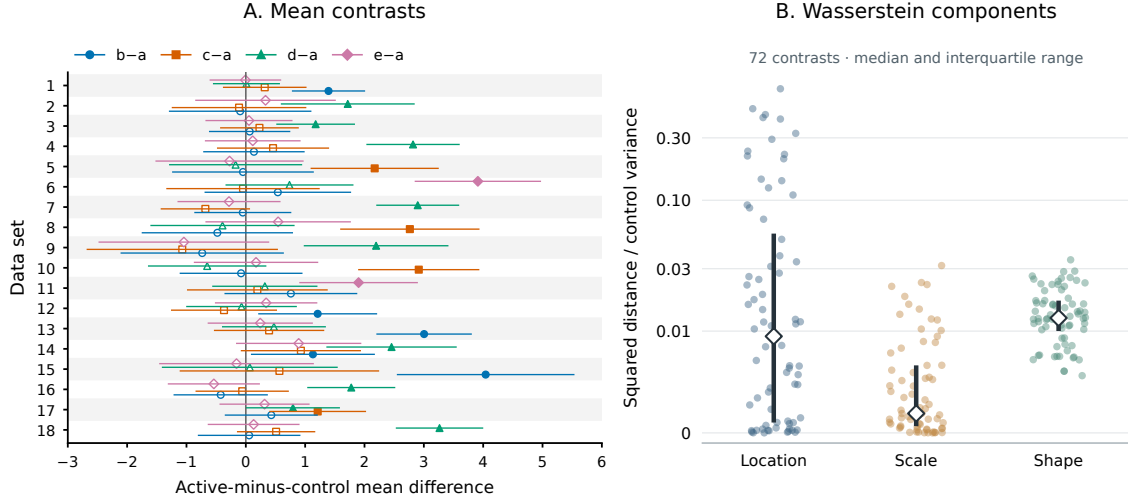


Figure 1: Outcome differences across 18 data sets. A: mean contrasts with pointwise 95% intervals; open markers indicate intervals containing zero. B: location, scale, and shape components of squared Wasserstein distance, divided by each data set’s control variance. Diamonds and bars show medians and interquartile ranges across 72 contrasts. The vertical scale is linear below 0.01 and logarithmic above.

We integrate over empirical quantile steps, using variance denominator n_j , and divide all components by the data set’s control variance. Location then accounts for 78.9% of the summed distance. Yet its median share within a contrast is only 29.9%, and it is the largest component in 31 of 72 contrasts.

These summaries answer different questions. Summing distances gives more weight to comparisons with large discrepancies. A few large mean differences can therefore dominate the total. The median instead describes a typical comparison, including those with small mean differences. In such comparisons, shape can account for a larger share even when its absolute contribution is small.

Empirical shapes also differ because each arm contains a finite sample. To gauge this variation, we pooled arm-standardized outcomes and drew 1,999 replicates per data set. We retained arm sizes, restored fitted arm means and scales, and re-estimated all terms, sharing one control sample across contrasts. Only three shape terms exceeded their pointwise reference 95th percentiles. Most observed shape differences are thus not unusually large under this fitted common-shape reference; this does not establish equal population shapes.

These comparisons describe outcomes averaged over covariates. They do not show whether a treatment works differently for different covariate values. They also provide background for the creator’s simulation targets, rather than a check of its fit: generated and observed outcomes are not compared here.

3.3 Covariate information for average-effect estimation

Can covariates explain outcome differences among units receiving the same treatment? If so, accounting for those differences may help estimate average effects. Randomization balances covariates on average, but the realized groups can still differ by chance. When those covariates predict outcomes, this imbalance adds noise to an unadjusted mean difference.

We fit a common-slope ANCOVA by ordinary least squares, using five arm intercepts and 57 shared covariate slopes $\hat{\beta}$. Let $\tilde{X}_i$ denote the centered, encoded covariates. The adjusted contrast compares the fitted arm intercepts. The normal equations express it as

$$\hat{\Delta}_z^{\text{adj}} = \bar{Y}_z - \bar{Y}_a - \hat{\beta}^\top (\widetilde{X}_z - \widetilde{X}_a).$$

The last term estimates how much of the observed mean difference is associated with covariate imbalance. Adjustment subtracts that contribution. Shared slopes keep the fit manageable in small arms, but may miss differences in how covariates predict outcomes across treatments (Zhao and Ding, 2023).

We use HC3 standard errors (MacKinnon and White, 1985). All 72 adjusted standard errors are smaller than the unadjusted ones, with a median reduction of 24.5%. This suggests that covariates may be useful for average-effect estimation. It is a comparison of estimated uncertainty, not a repeated-sampling demonstration of precision or coverage gains, and it does not measure out-of-sample prediction.

Why does useful outcome prediction not imply predictable effect variation? Suppose $m_j(x) = g(x) + c_j$. Covariates can strongly predict outcomes through $g(x)$, but the treatment contrast is the constant $c_z - c_a$: the same predictive contribution cancels. More generally, errors in two fitted outcome functions may cancel or reinforce when they are subtracted. Accurate outcome prediction alone therefore does not ensure accurate effect prediction (Künzel et al., 2019). This explains why the original procedure used outcome screening and a separate comparison against simulated effects.

4. Released truth and discussion

Released truth allows us to ask whether the outcome patterns above accompany actual variation in conditional effects. In these samples, iCATE is constant across units within each of the 72 contrasts. This information comes from the released effects, not from the preceding analyses.

We evaluate saved submission 0002.03 without refitting, matching units and treatments to released truth. Averaging data-set RMSEs equally gives 0.293 for PATE, 0.291 for sCATE, 0.300 for subCATE, and 0.618 for iCATE. The submission predicts within-contrast variation absent from the released effects. This comparison does not isolate the creator, learners, or selection as its source.

The observed data help explain variable representation, treatment-arm differences, and the role of adjustment. Released truth answers the separate question of whether effects vary across units. Plasmode construction itself can also distort comparisons, making some estimators appear overly biased (Shaw et al., 2026). Conclusions concern this curated subset; its constant sampled iCATE cannot assess recovery of genuine heterogeneity elsewhere in the challenge.

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